



Process & Flow Technology
8017 Breen Road
Houston, TX 77064
Telephone: (281) 351-2222

CUSTOMER: DNOW LP

P.O.# 662682068

S.O.#: 488071-1

PER YOUR INSTRUCTIONS, WE ARE ENCLOSING THE FOLLOWING DOCUMENTS:

☐	COPIES OF APPROVED DRAWINGS
☐	COPIES OF CERTIFIED DIM. DRAWINGS
☒	COPIES OF PARTIAL DATA REPORT
☒	COPIES OF MILL TEST REPORTS
	(ONE COPY ON CAP & ONE COPY ON HUB)
☐	COPIES OF SPARE PARTS LIST
☐	COPIES OF WELDING INSTRUCTIONS
☐	COPIES OF MAINTENANCE & OPERATIONS INSTRUCTIONS
☐	COPIES OF ASME CERTIFICATIONS
☐	COPIES OF DOT CERTIFICATIONS
☐	COPIES OF NACE CERTIFICATIONS
☐	COPIES OF API CERTIFICATIONS
☐	COPIES OF NDE CERTIFICATIONS
☐	



S.O. No. 488071-1

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An AFG Holdings Company

Material Test Report

Heat Code: JH2

JH2

AmeriForge Flange Products

13770 Industrial Rd. Houston, TX 77015

ISO 9001:2015 Certified

Sales: (713) 393-4200

NOV-R&M ENERGY SYS	15278	PO: 4046947	Sales Order: 198451	Line: 3
500 CONROE PARK WEST DR		Item Code: 9999H61985-0031R	Qty Shipped: 200	
CONROE, TX 77303		Item Desc: 04.00 150-900 HUB A350LF 17518507-001 NRM CDFRG		
		Mill Heat: C241468		
Spec: ASME SA350/SA350M(21) LF2 CL1 & NOV 61-0072 Rev 01 (Hubs)				

Element	(%wt)	Ladle	Product	EPCRA	CAS#	Element	(%wt)	Ladle	Product	EPCRA	CAS#
C	Carbon	0.18	0.18			Cr	Chromium	0.09	0.10	Y	7440-47-3
Mn	Manganese	1.13	1.17	Y	7439-96-5	Mo	Molybdenum	0.02	0.02		
P	Phosphorous	0.011	0.012			V	Vanadium	0.03	0.03		
S	Sulphur	0.01	0.01			Cb	Columbium	0.00	0.00		
Si	Silicon	0.23	0.24			Al	Aluminum	0.02	0.02		
Cu	Copper	0.13	0.13	Y	7440-50-8	C.E.		0.41			
Ni	Nickel	0.05	0.05	Y	7440-02-0						

Mechanical Testing		Heat Treat	
Test Lab	AFG	Norm. Temp. (F)	1675
Test Bar Size	4" 150-900	Norm. (Hrs)	2 HRS 30 MIN
HBW	146 - 146	Norm. Cooling Type	AIR
Elg (%)	33		
RA (%)	58		
Tensile Specimen Size (in)	.246	Other	
Tensile (ksi)	75.3	Country of Melt	USA
Yield (ksi)	45.5	Country of Forging	USA
Gauge Length (in)	.246	Vacuum Degassed	Yes
Specimen Orientation	LONGITUDINAL	Fully Killed	Yes
LCVN Charpy Temp. (F)	-50	Made to Fine Grain Practice	Yes
LCVN (ft lbs)	34 - 41 - 24		
LAvg 3(ft-lbs)	33		
L Min (ft-lbs)	24		
% Shear	20 - 20 - 10		
Lat. Exp.	.031 - .04 - .022		
		Approved for Manufacture	
		by	
		Date	7-25-24
			4" Hub

The recording of false, fictitious, or fraudulent statement or entries on this document may be punishable as a felony under federal statute.

EPCRA Supplier Notification: This product may contain one or more toxic chemicals subject to the reporting requirements of Section 313 of the Emergency Planning and Community Right-to-Know Act (Title III of the Superfund Amendments and Reauthorization Act of 1986) and 40 C.F.R. Part 372. Potentially reportable chemicals are indicated with a checkmark in the "EPCRA" column and a Chemical Abstract Services (CAS) registry number is provided for each such chemical in addition to the percent by weight of the chemical present in this product. It is your responsibility alone to determine whether your facility is required to submit a Toxic Release Inventory Report under EPCRA Section 313.

Certification No.: 1665670
Certification Date: 7/23/2024
Issued By: Michelle Simien

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This report is issued in compliance with the requirements of EN10204 3.1 / ISO 10474 3.1.b

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		Mill Heat: C241468		
Spec: ASME SA350/SA350M(21) LF2 CL1 & NOV 61-0072 Rev 01 (Hubs)				

TAG H61985

Product compliant with NACE MR0175/ISO 15156-2:2015, Annex A (A.2.1.2, A2.1.3).

Product compliant with NACE MR0103/ISO 17945:2015, Section 13.1.1 and 13.1.3.

- * No Weld repair performed.
- * Ladle Chemical Analysis results are reported from the raw material suppliers MTR.
- * Tensile taken from a round specimen. Measurement of elongation after fracture.
- * Yield strength was determined using the .2% offset method, unless specified.
- * All material supplied under this order is certified to be free of mercury contamination and no mercury bearing equipment was used in manufacturing, fabrication or testing.
- * When reported on MTR: Tensile Testing Per ASTM E8; Brinell Hardness Per ASTM E10; Charpy Testing Per ASTM E23
- * AF Global manufactures materials according to a certified Quality Management System conforming to ISO 9001:2015 and PED certified to 2014/68/EU, Annex 1, Section 4.3.
- Unless otherwise noted on PO, Dimensions of all standard pipe flanges and flange fittings (NPS 1/2 to NPS 24) per ASME B16.5-2020, large diameter steel flanges (NPS 26 to 60) per ASME B16.47-2020, Orifice flanges per ASME B16.36-2020.
- * AFGlobal uses the latest edition of ASTM E29 (Standard practice for using significant digits in test data to determine conformance with specifications), to claim conformance or nonconformance to a specification. When a accept/reject determination is required AFGlobal uses a test uncertainty ratio (TUR) of 1:1 or higher as acceptable.
- * DNV PED Certificate No 10000425664-PA-DNV-USA Valid thru December 03, 2026.

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Dusty Bannon - Quality Analyst

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